Sample-and-hold governance: review intervals as a design variable in resource stability

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October 16, 2026

Abstract

Institutions observe renewable resources at discrete times, revise rules at scheduled reviews, and hold commands between them. We model this architecture as sample-and-hold governance and distinguish it from both continuous-delay feedback and an annual-step approximation. We prove forward invariance of the nonnegative state space and finite-horizon consistency as the review interval tends to zero; neither result implies stability at any fixed positive interval. On an illustrative logistic core, the exponential frozen-assessment update has one unit-circle crossing at 6.501 years, whereas a forward-Euler command update reports crossings at 47.536 and 79.143 years. The exponential protective comparator is stable on the tested interval grid. The 6.501-year value is not a calibrated institutional threshold: across six-parameter joint perturbation boxes, a crossing is found at 60 of 64 corners at $\pm 0.5\%$ and 44 of 64 at $\pm 1\%$; every no-crossing corner is stable on a 20,001-point scan. In a 42-stock screen, the Benjamini–Hochberg-adjusted target-band count is zero. A recovered review-interval proxy test has only two proxy values, both below the modeled response windows, and its negative association is opposite the predicted approach gradient and confounded by region. A 32-system case inventory identifies no candidate meeting all four eligibility criteria. Northern cod contributes a descriptive two-window partition, not an identified governance mechanism. The results support operator-specific stability analysis, not a general claim that a particular review cadence stabilizes real fisheries.

Keywords: sample-and-hold control; review interval; resource governance; sampled-data stability; fisheries; management strategy evaluation.

Evidence status. The analytic propositions, deterministic multiplier calculations, archived trajectory classifications, empirical screens, and prospective designs have distinct evidential status. Supplementary Table S1 inventories the claims and limitations; parameter values in the numerical core are illustrative and uncalibrated.

1 Introduction

Renewable-resource institutions operate on schedules. Surveys generate observations; assessments translate observations into advice; a rule sets a catch or effort command; and that command is held, or updated only at the next authorized review. In fisheries, the review interval may be annual or multi-year, while the ecological state evolves continuously between decisions (Butterworth and Punt, 1999; Butterworth, 2007; Punt et al., 2016). The institutional clock is therefore part of the dynamical system, not merely a reporting convention.

Two common mathematical substitutes discard that architecture. A continuous-delay equation lets feedback act continuously on a lagged signal. An annual difference equation compresses all between-review dynamics into one step. Neither is generally the map implemented by an institution that samples, decides, and holds. Sampled-data control has long treated the zero-order hold as an

operator with its own spectral properties (Chen and Francis, 1995; Åström and Wittenmark, 1997; Nešić and Teel, 2004). Resource-governance models do not always preserve that distinction, even though stability can change when the decision operator changes.

A second problem is causal promotion: a spectral peak, cohort cycle, or crisis-driven monitoring record does not by itself identify the sign, timing, or responsiveness of a governance rule. The skeptical re-analysis of the Russell Cycle, for example, found that a long record treated as periodic did not resolve a true periodicity (McManus, Licandro, and Coombs, 2016). Here the same diagnostic discipline is applied to claims about institutional feedback. Periodicity is not treated as a mechanism label.

This paper makes four contributions. First, it specifies a sample-and-hold model in which observation, assessment, review, decision, deployment, ecological-response, and memory times are kept separate. Second, it proves forward invariance of the nonnegative state space and states precisely what the rapid-review limit does establish: finite-horizon approximation under regularity conditions, not finite-interval stability. Third, it compares an exponential frozen-assessment controller update with forward Euler on a common illustrative ecological plant and records how crossings move with the operator and parameter vector. Fourth, it re-examines the empirical layer: a multiplicity-controlled 42-stock spectral screen, a documented but coarse review-interval proxy test, a curated 32-system case search, and the northern-cod record.

The focal controller is deliberately diagnostic rather than normative. It raises extraction effort in response to a perceived shortfall, representing a mobilising or extractive feedback. A protective comparator reduces effort when the perceived deficit rises. The numerical coefficients are inherited illustrative conventions, not field estimates; the exercise therefore identifies operator dependence within a specified model class, not a calibrated review interval for a real fishery. This distinction matters especially because the baseline crossing lies close to the unit circle and moves substantially under small parameter perturbations.

The paper proceeds as follows. Section 2 defines the model and evidence procedures. Section 3 reports the two mathematical properties, the operator-specific crossing record, sensitivity, and empirical results. Section 4 states what transfers to management design and what remains unidentified. Detailed proofs, case records, data tables, and computational audits are in the Supplementary Information.

2 Materials and methods

2.1 Sample-and-hold model

Let review times be $t_n = nT_\mathrm{r}$, where T_r is the time between opportunities to revise a management command. Raw observations and decision opportunities need not share a cadence. The seven time objects used here are kept distinct (Table 1).

Table 1: Institutional and ecological time objects. The quantities are not collapsed into one governance lag unless the data cannot resolve them and an aggregation rule is stated.

Object	Definition
Observation interval T_obs	Time between raw measurements.
Assessment interval T_assess	Time between formal state estimates.
Review interval T_r	Time between opportunities to change the command.
Decision lag τ_dec	Assessment completion to formal decision.
Deployment lag τ_dep	Decision to implemented change in extraction pressure.
Ecological response lag τ_eco	Implementation to detectable ecological response.
Memory timescale τ_m	Relaxation time of a filtered institutional signal; not a discrete delay.

In fisheries language, the controller is a harvest-control rule, its command is the catch or effort limit, the review interval is the advice cycle, and the hold is the period between revisions. Consider

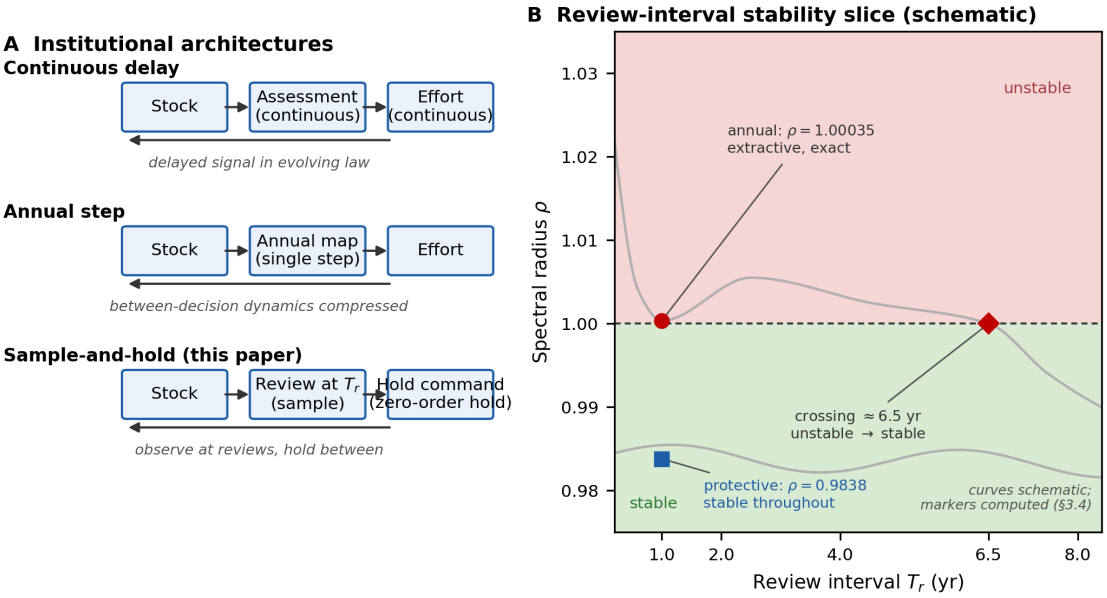


Figure 1: Three representations of institutional timing. Continuous-delay feedback acts on a lagged signal; an annual-step model compresses the between-decision dynamics; sample-and-hold governance observes and revises at review times and holds the command between them. Panel B is schematic: the computed markers identify the annual spectral radii and the exponential-update crossing, not a universal institutional response curve.

a resource stock $N(t)$, a nonnegative institutional deficit signal $Z(t)$, and effort E_n held fixed on $[t_n, t_{n+1})$. The plant and signal evolve as

$$\dot{N}(t) = rN(t)\left(1 - \frac{N(t)}{K}\right) - qE_nN(t), \tag{1}$$

$$\dot{Z}(t) = \frac{\Phi_k(qE_nN(t) - S(N(t))) - Z(t)}{\tau_m}, \qquad S(N) = rN\left(1 - \frac{N}{K}\right), \quad \tau_m > 0. \tag{2}$$

The signal map is a shifted, floored softplus,

$$\Phi_k(s) = \max\left\{0, \frac{1}{k}\log(1 + e^{ks}) - \frac{\log 2}{k} + \delta\right\}, \qquad \delta = \frac{\log 2}{10}, \quad k = 10.$$

At each review the contemporaneous end-of-interval assessment $\widehat{Z}_{n+1} = Z(t_{n+1}^-)$ is read. The mobilising effort law below defines a controller rate, and two command-update operators are compared:

$$E_{n+1}^{\exp} = \Pi_{[0,E_{\max}]}\left(\varphi_{T_r}^{F_B}(E_n, \widehat{Z}_{n+1})\right), \tag{3}$$

$$E_{n+1}^{\text{Euler}} = \Pi_{[0,E_{\max}]}\left(E_n + T_rF_B(E_n, \widehat{Z}_{n+1})\right), \tag{4}$$

$$F_B(E, Z) = \left(1 - \frac{E}{E_{\max}}\right)\left[\eta E\left(\frac{Z}{\Delta_{\text{ref}}} - \frac{E}{E_{\max}}\right) + \delta_0\frac{Z}{Z_{\text{ref}} + Z}\right]. \tag{5}$$

Here $\varphi_T^{F_B}(E, \widehat{Z})$ is the time-T flow of $d\mathcal{E}/ds = F_B(\mathcal{E}, \widehat{Z})$, initialized at $\mathcal{E}(0) = E$, with the endpoint assessment held fixed during the controller update. Thus the exponential update exactly integrates the frozen-assessment controller ODE; it is not an exact solution of the coupled ecological-governance system. The Euler update is its one-step forward-Euler alternative. In both cases, the plant and signal have first evolved for T_r with effort E_n held constant. The law is a declared illustrative controller, not a generic harvest-control rule. Since the command update uses T_r, changing the

review interval changes both the hold duration and the controller-update horizon. The same-plant protective comparator is

$$F_B^{\mathrm{prot}}(E,Z)=\left(1-\frac{E}{E_{\max}}\right)\eta_p\big(E_{\mathrm{cap}}(Z)-E\big),\qquad E_{\mathrm{cap}}(Z)=E_0\frac{Z_{\mathrm{ref}}}{Z_{\mathrm{ref}}+Z},$$

with $\eta_p=\eta$ and $E_0=E^*(Z_{\mathrm{ref}}+\delta)/Z_{\mathrm{ref}}$, so the interior rest coincides with that of the extractive rule. Projection, flow, and update ordering are fixed as written; reversing the order defines another hybrid system.

For the local stability calculation, let $e_n=E_n-E^*$, $\hat{z}_{n+1}=\widehat{Z}_{n+1}-Z^*$, and $C_E=\partial_E F_B(E^*,Z^*)$, $C_Z=\partial_Z F_B(E^*,Z^*)$. Linearizing the exact frozen-assessment controller flow and the Euler update gives, respectively,

$$e_{n+1}^{\exp}=e^{C_E T_{\mathrm{r}}}e_n+\frac{e^{C_E T_{\mathrm{r}}}-1}{C_E}C_Z\hat{z}_{n+1},\tag{6}$$

$$e_{n+1}^{\mathrm{Euler}}=e_n+T_{\mathrm{r}}\left(C_E e_n+C_Z\hat{z}_{n+1}\right).\tag{7}$$

The quotient has its continuous limit T_{r} when $C_E=0$. In the exponential case this is the exact local flow of $\dot{e}=C_Ee+C_Z\hat{z}_{n+1}$ with the endpoint assessment fixed; it is not the exact nonlinear controller flow written in closed form. The full monodromy composes this post-flow command update with the held-plant flow, so the dependence of $\hat{z}_{n+1}$ on the pre-review state is included. The review map is denoted $\mathcal{P}_{T_{\mathrm{r}}}$, with monodromy $D\mathcal{P}_{T_{\mathrm{r}}}(X^*)$. Local asymptotic stability requires all its eigenvalues to lie inside the unit disk.

The illustrative numerical baseline vector is:

$$(r,K,q,\eta,E_{\max})=(0.02,100,0.001,0.914,30),\qquad(\delta_0,Z_{\mathrm{ref}},\Delta_{\mathrm{ref}},\tau_m)=(0.01,1,1,5),$$

in model units, with $Z^*=\delta$. It gives $(N^*,Z^*,E^*)=(89.55188,0.069315,2.08962)$ and $F^*=qE^*=0.002090\,\mathrm{yr}^{-1}$. The institutional coefficients are not identified from field data; $\eta=0.914$ is an inherited illustrative baseline. The full vector, equilibrium derivation, gains, and stage-map conventions are in Supplementary S8 and S11.

2.2 Structural results and numerical protocol

The admissible state set is $N\geq 0$, $Z\geq 0$, and $E\in[0,E_{\max}]$. The signal map is nonnegative and effort commands are projected onto the admissible interval. Supplementary Proposition S2.1 establishes forward invariance for held or convex-interpolated effort. Separately, if the continuous vector field is C^1 with bounded derivative on a compact neighborhood of compared trajectories, assessment is contemporaneous, projection is locally inactive, and there are no additional queues or stochastic updates, the flow-then-update map has local one-step defect $O(T_{\mathrm{r}}^2)$. Discrete Gronwall then gives uniform $O(T_{\mathrm{r}})$ error on each fixed finite horizon as $T_{\mathrm{r}}\to 0$. This is a finite-horizon approximation statement, not a finite-interval stability theorem.

For the logistic core, the linearised held-flow matrix is combined with the exponential frozen-assessment or Euler review update; its spectral radius $\rho(T_{\mathrm{r}})$ is scanned over $[0.2,200]$ years on a 200,001-point grid, with bisection refinement at sign changes. The crossing scan was re-executed against the deposited campaign and agrees with the archived crossing table. A separate high-resolution sensitivity implementation evaluates the six-parameter joint boxes on 20,001 review-interval points. Numerical scans are not presented as analytic global stability theorems. A finite-difference audit of the full nonlinear plant plus frozen-assessment controller flow reproduces the local Jacobian and crossing; details and limitations are given in Supplementary S11. No invariant-circle continuation was completed.

A second plant, a delayed-recruitment two-stage map, is included only as a distinct structural comparison. The archived trajectory-classified windows are provisional because their original solver configuration and histories are not attached. The pre-registered stage-map reconstruction is a newly

specified object: its comparison with those windows is a consistency check, not a validation. The state dimension, plant, and effort scale differ from the logistic core, so cross-plant response regions do not isolate an operator effect; that contrast is made only on the common logistic plant.

2.3 Empirical and case procedures

Stock screen and power. The input is the RAM Legacy Stock Assessment Database v4.66. Of 58 forage-fish stock series, 42 have at least 20 non-missing annual spawning-stock-biomass (SSB) points (range 20–77 years). The analysis linearly detrends each SSB series, computes a unit-total-power Lomb–Scargle periodogram on 1,500 frequencies, and integrates power in bands A (2.5–5 years) and B (5–9 years). For each stock and band, an AR(1) null is fitted to the detrended series and simulated 200 times with seed 7; empirical $p = (k + 1)/201$ values are adjusted by Benjamini–Hochberg over the 84 stock-by-band tests. These are SSB periodicity diagnostics, not controller-sign tests. The annual-review label and the review-interval proxies are analyst assignments, not database fields or per-stock primary-verified classifications. Sensitivity variants, end-point trims, and alternative bands are reported in Supplementary S9.

The separate T_{r}-proxy test uses the group-documented coding in the recovered script: seven Peru/Chile stocks receive $T_{\mathrm{r}} = 0.5$ years (annual quota with in-season responsive closures), and 35 annual-TAC proxies receive $T_{\mathrm{r}} = 1$ year. The 4-year anchovy and 8-year sprat response windows are both above these two proxy values. Thus the cohort contains no documented multi-year responsive review interval with which to estimate a window-approach gradient. The proxy assignments are coarse and not primary-verified; the seven short-interval cases are geographically concentrated, which limits any cross-sectional interpretation.

A separate injected-signal exercise runs the committed sampled-governance effort model with multiplicative lognormal observation noise. The canonical white-noise table comprises 17 cells, 50 trials per cell, trial seeds 0–49, and a 120-replicate seed-7 AR(1) detection null. It quantifies conditional detection under the specified simulation grid; it is not a per-stock minimum-power guarantee.

Structured case search. The 32-system inventory is curated rather than systematic. Each candidate is assessed against four requirements: (i) responsive governance rather than a one-time cap or ban; (ii) an independently dateable response or implementation lag; (iii) no competing ecological or structural driver supported for the same outcome; and (iv) individual-resource or station-level data. The evidential rule for (iii) is comparative: institutional feedback must outperform prespecified alternatives on held-out prediction, phase order, or intervention response. The search log and full row-level dispositions are in Supplementary S4.

Northern cod. The northern cod (NAFO 2J3KL) record is treated as an assessment-based descriptive case. The strong-Allee surplus equation and its phase-line result are model-class diagnostics, not a fitted explanation of the stock trajectory. Assessment-table values from DFO CSAS SAR 2016/026 are used to distinguish the 1991–1995 crash window from the later non-recovery record. Mortality allocation differs across assessment formulations, and the constrained-mortality quantities are not reproduced here; they remain hypotheses and open reconstruction requirements (Supplementary S14.2).

3 Results

3.1 Invariance and the scope of the rapid-review limit

Under the admissibility conditions in Section 2.2, the interval $N \geq 0$, $Z \geq 0$, $0 \leq E \leq E_{\max}$ is forward invariant. The logistic boundary has zero population growth at $N = 0$; the signal derivative cannot point outward through $Z = 0$ because its forcing is nonnegative; and projection keeps effort in its prescribed bounds. This is a state-space result, not a stability result.

As $T_r \to 0$, the held flow and contemporaneous update converge on fixed finite horizons to the corresponding continuous-time system when the vector field is regular and no nonsmooth boundary is crossed. For a fixed positive T_r, local stability is decided by the eigenvalues of the complete review map, including the plant flow and update law. It is therefore possible for the rapid-review approximation to be accurate at short horizons while a finite-interval map has a different stability classification. The formal statement and proof are in Supplementary S2.

3.2 Response windows and operator-specific crossings

The archived trajectory-classified stage model reports persistent oscillation near a 3–4-year anchovy review interval and across a 6–12-year sprat interval for its baseline parameterisation. These are provisional classifications: the original solver and initial histories were not deposited. A newly specified, preregistered stage-map reconstruction did not reproduce those anchovy or sprat oscillation criteria; it did reproduce annual-review convergence for anchovy and convergence for the cod and slow-stock criteria. This mismatch is not averaged away: the reconstruction and archived classification are different evidence objects, and a mismatch does not adjudicate the unavailable legacy generating object (Supplementary S9.2).

On the common illustrative logistic plant, the exponential frozen-assessment mobilising update has a single complex-pair unit-circle crossing at $T_r = 6.50129495$ years. The controller update exactly integrates the effort-rate ODE with the endpoint assessment fixed; the multiplier calculation uses its local linearization. The crossing angle is 0.175494 radians, or about 0.05586π. The sampled multipliers are outside the unit circle immediately below the crossing and inside immediately above it. The forward-Euler implementation, on the same plant and baseline vector, has a complex-pair crossing at 47.535982 years and a second, real -1 crossing at 79.142688 years. Thus the transition pattern is update-operator dependent; the Euler values are not estimates of an institutional threshold. A continuous-delay comparator from the companion analysis gives crossings near 3.666 and 150.358 years (Abaee, 2026), further illustrating that a continuous lag, a frozen-assessment controller update, and a forward-Euler command step are different models (Fig. 2; the common-plant scope is detailed in Supplementary S11).

The exponential frozen-assessment protective comparator has no crossing on the tested $[0.2, 200]$-year grid, with maximum sampled spectral radius 0.996740. This numerical statement is specific to the stated gain convention, equilibrium, projection-inactive local map, and grid. It is not a general theorem that protective policies are stable.

Table 2: Baseline local stability record on the illustrative logistic plant. A crossing is a numerical unit-circle crossing of the specified map; it is not a field-estimated review interval.

Controller/update	Computed transition	Interpretation
Mobilising, exponential frozen-assessment	6.501295 yr; one complex pair	Unstable below, stable above the crossing on the scanned branch.
Mobilising, forward Euler	47.535982 yr and 79.142688 yr	Complex-pair crossing, then real -1 crossing; distinct Euler map.
Protective, exponential frozen-assessment	No crossing on $[0.2, 200]$ yr; max $\rho = 0.996740$	Stable on the tested grid only.
Continuous-delay comparator	About 3.666 and 150.358 yr	Different continuous-lag operator; shown for contrast.

3.3 Parameter sensitivity and joint perturbations

The baseline exponential-update crossing is fragile to some scalar parameters. In the archived one-at-a-time $\pm 5\%$ battery, increasing K, q, or $E_{\max}$ can move the crossing to about 13.8–14.3 years, while decreasing these parameters can remove the crossing from the tested interval; perturbing η produces crossings from about 2.19 to 8.72 years. Changes to the offset d_0 move the crossing

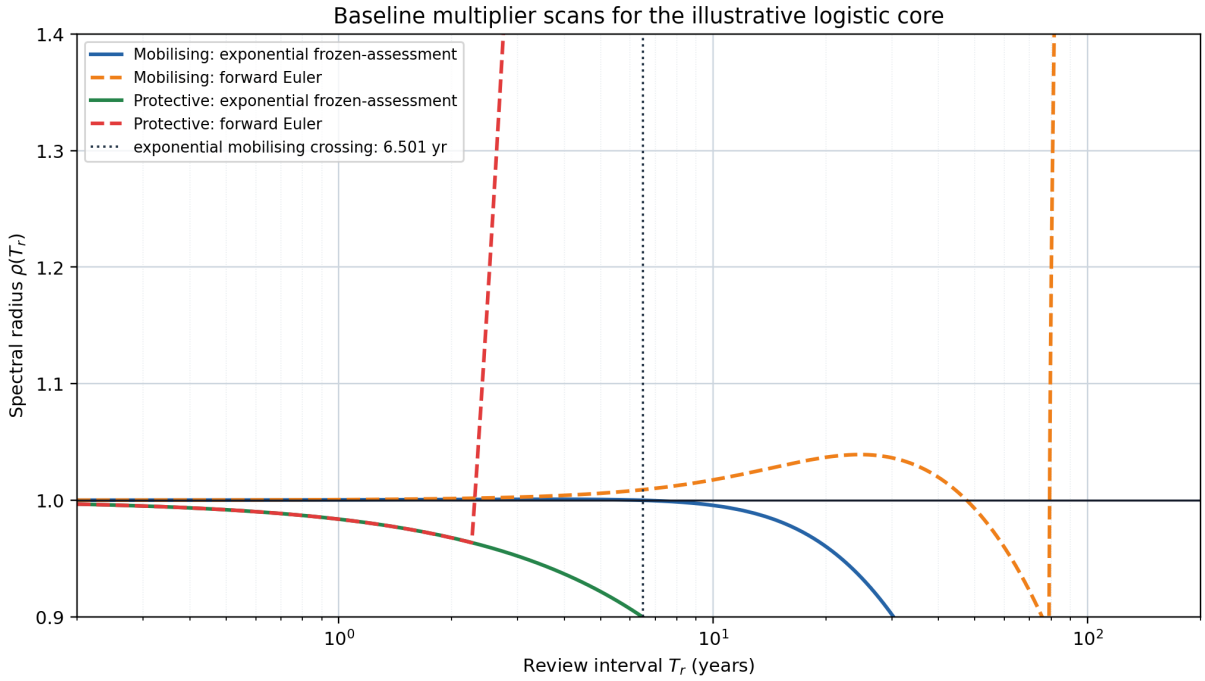


Figure 2: Baseline spectral-radius scans for exponential frozen-assessment and Euler mobilising and protective maps. The exponential mobilising crossing is near 6.50 years; the plotted vertical range is restricted to $0.9 \leq \rho \leq 1.4$ for readability. Curves are model outputs, not empirical response curves.

only modestly, from about 6.47 to 6.53 years, while the reference scale and memory time have smaller but non-negligible effects. The crossing-only summaries of the archived ±5% and ±10% battery are tabulated in Supplementary S11.4; the full archived rows and fields are retained in `src/p7_empirical/recovered/sens_table.csv`. The separate rerun ±2% crossing table is in Supplementary S11.3.

A separate joint-box analysis perturbs six parameters $(r, K, q, \eta, E_{\max}, \Delta_{\text{ref}})$ simultaneously. At ±0.5%, 60 of 64 corners have a crossing, with the first-crossing values spanning 0.871965–10.670425 years; at ±1%, 44 of 64 have a crossing, spanning 4.423192–13.653242 years. All 4 and 20 no-crossing corners, respectively, remain inside the unit circle at every point on the 20,001-point review-interval grid. The largest sampled radii among these no-crossing corners are 0.9999713 and 0.9999953. These are dense-grid numerical classifications, not proofs of stability between grid points, and the perturbation boxes are not confidence regions. They show that the 6.50-year baseline value should not be read as a robust threshold.

3.4 Forage-fish screen: no multiplicity-adjusted target-band detections

Among 42 RAM Legacy v4.66 forage-fish stocks with at least 20 annual SSB points, no stock-band cell is significant after Benjamini–Hochberg adjustment across 84 tests in the baseline 2.5–5-year and 5–9-year bands. The smallest raw empirical p-value is 0.0448, but its adjusted value is 1.00. A wider sensitivity battery varying detrending, end-point trimming, red-noise simulation counts, seeds, and band definitions also yields no adjusted target-band detections; one nominal baseline band-B flag does not survive multiplicity control. The result is evidence about this particular diagnostic and cohort, not evidence that governance feedback is absent.

The screen's null-excess distribution has many values below its AR(1) 95% thresholds, but an AR(1) null controls only the fitted red-noise structure. It does not remove quasi-periodic external forcing, regime shifts, observation error, or assessment changes. In particular, the stock screen does not estimate the sign of a harvest-control rule or the causal effect of its review interval. A separate 17-cell synthetic power table detects sprat-class signals at 1.00 in the low-noise $H = 100, 200$

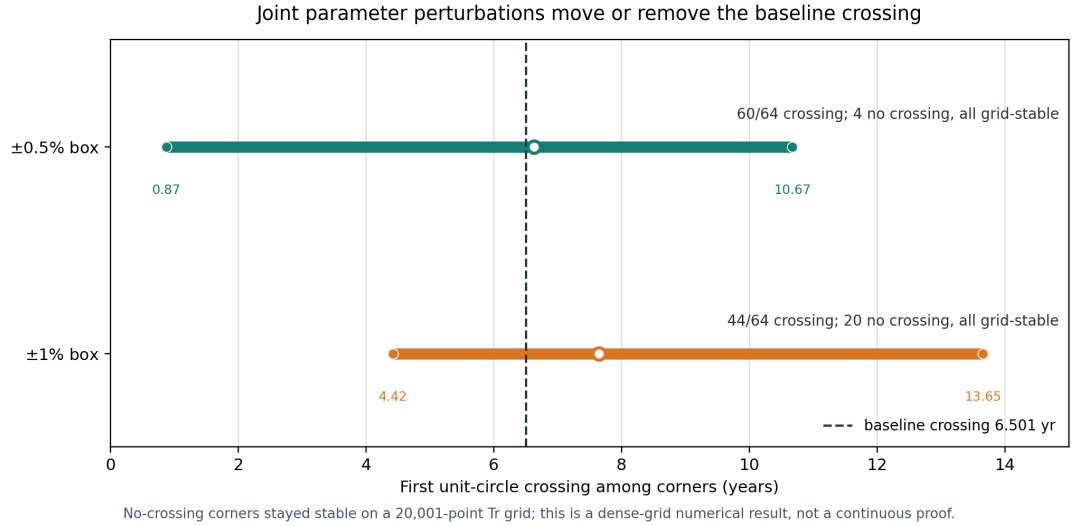


Figure 3: Six-parameter $(r, K, q, \eta, E_{\max}, \Delta_{\text{ref}})$ joint-box sensitivity of the exponential-update first crossing. Points show the minimum and maximum crossing among corners that cross; the annotated no-crossing corners were stable at all 20,001 tested review intervals. Boxes are deterministic perturbation sets, not probability or confidence intervals.

cells and at 0.24–0.70 in the corresponding high-noise cells; anchovy-class power is 0.00–0.10 and the cod-labelled false-positive rate 0.02–0.08. These are conditional simulation proportions, not per-stock power guarantees, and the table was assembled from archived outputs rather than a complete driver rerun (Supplementary S14.1). Fig. 4 displays the baseline adjusted values and stock-level null comparisons; full per-stock results and variants are in Supplementary S9.

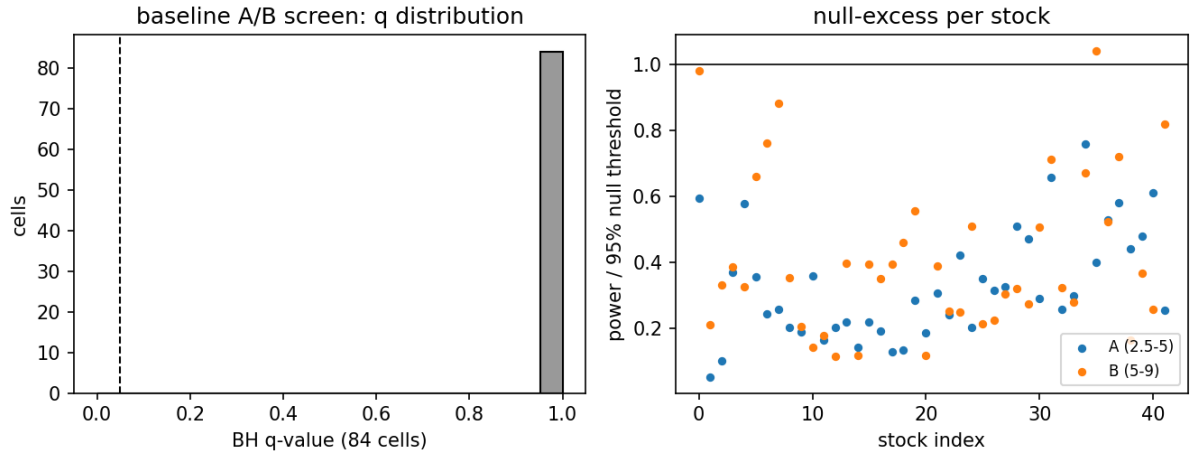


Figure 4: Baseline SSB periodicity screen. Left: Benjamini–Hochberg q-values for 84 stock-by-band tests; the dashed line denotes 0.05. Right: per-stock band power relative to the fitted AR(1) 95% null threshold (A: 2.5–5 years; B: 5–9 years). A high ratio is a diagnostic exceedance measure, not a causal governance signal.

3.5 Review-interval proxy test: no approach gradient is identifiable

The re-executed T_{r}-ranked test contains 42 stocks but only two assigned proxy values: seven Peru/Chile cases at 0.5 years and 35 annual-TAC cases at 1 year. Both proxy values lie below both sampled response windows (4 years for the anchovy-class window and 8 years for the sprat-class window). There are no documented responsive multi-year cases in this cohort from which to

estimate how spectral excess changes as T_r approaches a modeled window. The correct inference is therefore *no identifiable window-approach gradient*, not a supported gradient and not a direct empirical confirmation of a particular threshold.

The two-group contrasts do reproduce: the 0.5-year Peru/Chile group has greater median band-A null-excess than the annual group (0.529 versus 0.257; two-sided Mann–Whitney $p = 0.007$), and the rank association with assigned T_r is negative for band A (Spearman $\rho = -0.408, p = 0.007$) and band B ($\rho = -0.340, p = 0.028$). The sign is opposite to a simple approach-to-window prediction: the group assigned the shorter interval is farther from either modeled window. These reproduced two-group associations must not be described as an approach gradient or as an institutional effect. The 0.5-year assignments are group-level proxies for in-season responsive closures, not primary-verified stock-specific review intervals; all seven are Peru/Chile cases. Regional ecology, including ENSO-related quasi-periodic forcing, is a plausible confound, but is not established as the cause by this analysis. The AR(1) null absorbs fitted red noise, not quasi-periodic forcing. At nominal 95% thresholds, the test records 0/42 band-A flags and 1/42 band-B flag; these are not additional BH-adjusted discoveries. Per-stock intervals, thresholds, and coding notes are in Supplementary S10.

The companion F-placement audit locally reproduces the 42-stock cohort, matches fisheries-ADH F to last-F for 42 of 43 intersected stocks, and matches all 13 broader-pool taxonomic-group F quartiles. Mean stock-level F is associated with band-A null-excess before excluding Peru/Chile ($\rho = 0.3069, p = 0.0609$) but not after exclusion ($\rho = 0.0339, p = 0.8565$); band-B results show the same attenuation. This is consistent with regional confounding, not proof of its cause. The archived ADH formula and cohort summaries also reproduce, but ADH remains a calculated buffer-years proxy rather than a direct institutional-response measure (Supplementary S13.2).

3.6 Northern cod: a descriptive two-window record

The assessment record separates a 1991–1995 decline from the post-1995 low-stock period. SSB falls from about 730 kt in 1991 to 11 kt in 1995; the later assessment values remain low through the mid-2000s before increasing to about 300 kt in 2015. Natural mortality varies substantially across assessment years and is not a monotone proxy for governance. The window split is descriptive: it does not identify a switch in institutional timing, nor adjudicate environmental, predation, model, and management explanations. The constrained-mortality formulation supplies useful hypotheses but its underlying quantities are not reproduced in this edition. Fig. 5 and Supplementary S13.3 preserve the values and period definitions.

4 Discussion

4.1 The transferable result is an operator distinction, not a cadence threshold

The exponential frozen-assessment and forward-Euler controller updates, and the continuous-delay comparator, yield different spectral transitions on the same illustrative ecological plant. This is the central mathematical result. It cautions against transferring a stability boundary from one representation to another without matching (i) what is observed and when, (ii) which state estimate enters the rule, (iii) when the command changes, (iv) whether it is held between reviews, and (v) how command gain scales with the interval. In a management-strategy evaluation, those items should be explicit alongside operating-model uncertainty and implementation error (Punt et al., 2016).

The comparison does not show that real fisheries are unstable under annual reviews or that a longer interval is generally safer. The baseline exponential-update crossing is parameter-sensitive, the controller is uncalibrated, and the exponential protective result is only a scan over a finite interval grid. A fitted institutional threshold would require independent estimates of ecological response, observation and assessment error, decision and deployment lags, and the effective rule

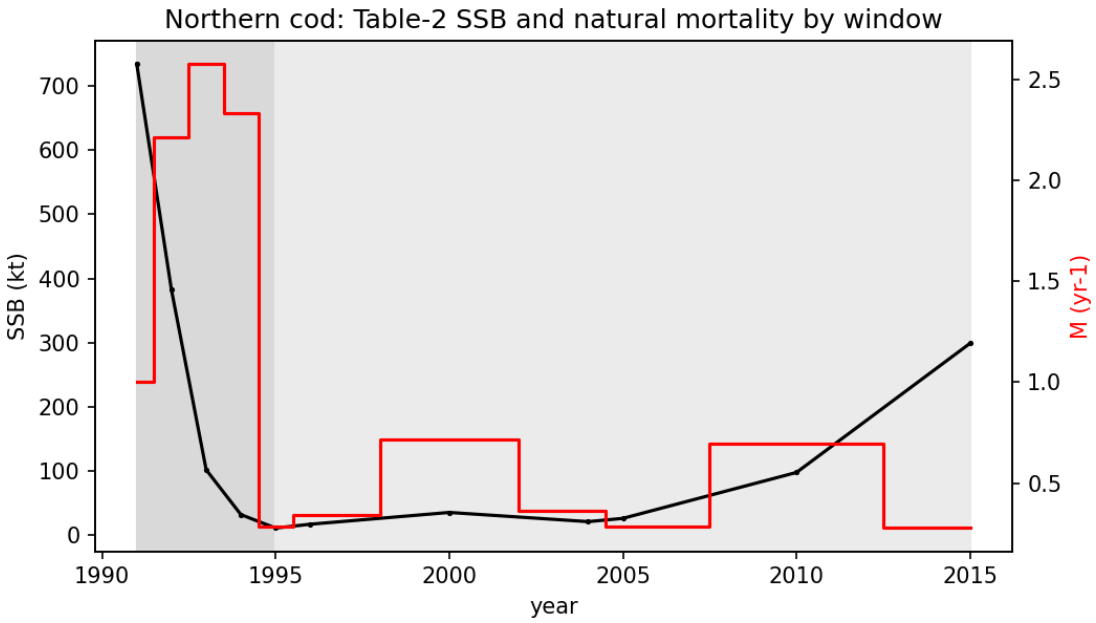


Figure 5: Northern cod assessment SSB and reported natural mortality by assessment year, with the 1991–1995 crash interval separated from the later record. The partition is descriptive and does not estimate a governance effect.

gain. Without them, the reported years are reproducible properties of a stated model, not policy prescriptions.

4.2 Empirical diagnostics constrain stronger claims

The baseline forage-fish screen contains no multiplicity-adjusted target-band detection, and the sensitivity variants do not change that decision. This is a useful negative diagnostic but not a test of a causal governance mechanism. Lomb–Scargle band power can be elevated by external forcing and structural change; an AR(1) null does not model all such alternatives. Nor does a spectrum alone reveal whether management responded to a signal before or after the ecological change.

The corrected reading of the review-interval test is particularly important. There is no observed interval spanning the modeled response windows, so there is no identifiable window-approach gradient. What is reproduced is an opposite-direction contrast between two proxy groups: seven regionally clustered 0.5-year assignments have larger null-excess than 35 annual assignments. Since the shorter proxy is farther from both modeled windows, this contrast does not support the proposed approach mechanism. The two-value coding, group-level rather than stock-level verification, and regional concentration leave confounding unresolved. ENSO-related quasi-periodicity is a candidate explanation, not a demonstrated cause. This distinction replaces any earlier shorthand suggesting that the cross-sectional test found a predicted gradient.

The 32-system search likewise has a narrow interpretation. No inventory candidate satisfies all four evidential requirements, but the inventory is a curated search rather than an exhaustive review, and missing station-level data could exclude otherwise relevant cases. The result is that a qualifying demonstration was not found in this search, not that institutional feedback cannot occur.

4.3 A descriptive cod partition is not a causal decomposition

Northern cod provides an instructive record because the collapse and subsequent low-stock period are well documented. However, a two-window plot cannot identify the causal contribution of harvest rules, predation, climate, recruitment, observation, or assessment structure. The strong-Allee surplus model is a local mathematical lens; unless fitted against alternatives and validated

out of sample, it does not establish an Allee mechanism. Likewise, the constrained-mortality formulation discussed in prior notes cannot support numerical conclusions here until its source table, uncertainty, and complete computation are recovered. The manuscript therefore retains only the reproducible descriptive partition and lists the unreproduced mortality claims as open hypotheses in Supplementary S14.2.

4.4 Requirements for a governance-feedback test

A stronger empirical design should treat institutions as measured mechanisms rather than inferred labels. At minimum, it would require:

1. dated observations, assessment releases, rule decisions, implementation, and ecological response at a resolution finer than the candidate response window;
2. stock-specific evidence about the operative rule, compliance, command persistence, and the timing of responsive revisions;
3. a state-space model that separates observation error, process noise, external forcing, regime shifts, and structural assessment changes;
4. pre-specified comparisons among no-feedback, delayed-feedback, sample-and-hold, and ecological-only models, with held-out predictive scoring or a credible intervention design; and
5. uncertainty propagation for the estimated review map, rather than reporting only a single point crossing.

These requirements are compatible with management-strategy evaluation and adaptive management practice (Walters, 1986; Punt et al., 2016), but are more demanding than detecting a spectral peak. Supplementary S7.1–S7.6 specifies five prospective designs, their decision rules, and cross-cutting preregistration requirements; none is reported as executed.

4.5 Limitations

This edition has six important limitations. First, the focal numerical parameters are illustrative, with no joint field calibration. Second, local eigenvalue stability does not imply global nonlinear stability or acceptable transient performance. Third, dense-grid non-crossing results are numerical evidence only. Fourth, the RAM screen and T_{r}-proxy assignments inherit database coverage and analyst coding; stock-specific primary verification is incomplete. Fifth, the stage-map reconstruction does not reproduce all archived trajectory windows, which remain provisional. Sixth, the case inventory and cod analysis do not identify a causal governance effect. Supplementary materials retain complete outputs and provenance so that these limitations are not hidden by a concise main text.

5 Conclusion

Review cadence is a dynamical design variable when decisions are sampled and commands are held, but a cadence threshold is meaningful only for a fully specified and empirically supported update map. In the illustrative system, exponential frozen-assessment, Euler, and continuous-delay operators produce different crossings, while small joint parameter perturbations move or remove them. The analytic invariance and rapid-review results do not supply a stability guarantee at finite review intervals. The empirical screens do not reveal a window-approach gradient; the reproduced two-group association points in the opposite direction and is confounded by coarse, regionally clustered proxies. Northern cod remains a descriptive case. These findings motivate operator-aware, data-rich evaluations rather than a general claim that any particular governance interval stabilizes real resources.

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Declarations

Funding No external funding was received for this work.

Conflicts of interest

The author declares no competing interests.

Data and code availability

The source bundle contains the model core, analysis scripts, archived tables, recovered local inputs used for re-execution, and the Supplementary Information. The RAM Legacy database is identified by version and source; upstream data remain subject to their original terms. A reconstruction register distinguishes newly reproduced outputs from archived records and unresolved inputs. The full 17-cell white-noise power table is assembled from the archived 15-cell run log and the two-cell cod follow-up JSON; a complete rerun of the driver did not finish. This limitation is stated in Supplementary S14.1.

Author contributions

A.A. conceived the analysis, developed the model, conducted the re-analysis, and wrote the manuscript.

Ethics approval

Not applicable.

AI-assisted tools

AI tools assisted with drafting, code generation, and iterative review. The author reviewed and edited the manuscript and takes responsibility for its content.